

Simulations of a Dense Plasma Focus on a High-Impedance Generator

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Abstract—Naval Research Laboratory is starting to study the connection between plasma instabilities and fast ion acceleration for neutron production on a dense plasma focus (DPF). The experiments will be performed on the HAWK generator (665 kA) that has a fast rise time, 1.2 μ s, and a high inductance, 720 nH. There is a conjecture that the neutron yield on a DPF may be enhanced on a high-impedance generator, since the assembly on the axis will not significantly reduce the current drive. Furthermore, to prevent the upstream breakdown during the implosion, there are plans to inject the gas load as plasma far from the insulator stack. As an introductory study, we performed MHD simulations using the Athena code. Athena is a Eulerian 3-D, unsplit finite volume MHD code that includes shock capturing with Riemann solvers and resistive diffusion for the magnetic field components. The simulations are coupled to an equivalent circuit model for HAWK. We report the dynamics and implosion time as a function of the initial injected plasma distribution and study the implications of nonideal effects. We also calculated a stochastic particle acceleration resulting from the motional electric field.

Index Terms—Ion accelerators, magnetohydrodynamics, neutrons, plasma devices, plasma simulation.

I. INTRODUCTION

A DENSE plasma focus (DPF) is a type of Z-pinch device where a Z-pinch is formed at the end of a cylindrical anode after an electric discharge creates a plasma between the cylindrical electrodes, and the magnetic field sweeps the plasma to the end of the anode (see [1]). The sweep-down phase is designed to delay the implosion so that the generator circuit can drive the maximum current when the plasma pinches. Research in DPFs began in the 1950s and 1960s and quickly gave a measurable flux of fusion neutrons, which

were identified as coming from beam-target reactions [2], [3], supposedly appearing after instabilities destroy the pinch and disrupt the current. Improvements in the pinch configuration, based on an empirical understanding, can lead to a significant increase in neutron yield as well as the maximum neutron energy [4]. Although the modeling of DPFs has been ongoing for four decades [5], so far a physical mechanism for ion acceleration has not been firmly established, which complicates further progress in the design and setup of DPFs.

The physics of ion acceleration in DPFs presents a paradox. While particle acceleration depends on nonideal electric fields, it is only manifested in experiments when the collapsed plasma is dense enough, typically $\gg 10^{18} \text{ cm}^{-3}$. However, at these densities, ideal MHD is typically an accurate model, and thus nonideal electric fields should not be expected to develop. Experimental and theoretical investigations into the development of these nonideal electric fields are the main focus of the new research effort in DPFs currently underway at Naval Research Laboratory (NRL).

The sweep up of the plasma and the collapse on the axis can be accurately described in terms of MHD dynamics until some particular time when current disruption occurs and strong electric fields are hypothesized to develop.

The purpose of this paper is twofold. The first goal is to provide the theoretical and numerical support for the DPF experiment on the NRL Hawk pulsed-power generator. For the description of Hawk see [6]. The DPF installed on Hawk is unusual in that it has local plasma injection via several plasma guns. Plasma is injected into the evacuated space between the coaxial electrodes. This is different from a typical DPF installation which has a neutral gas fill and breakdown at the insulator. The configuration of the DPF on Hawk, therefore, is not axisymmetric by design and this configuration should be modeled to see how it affects the formation of the pinch. It should be noted that the uniform gas fill will not provide a perfectly axisymmetric implosion either, due to the random nature of the initial breakdown. On Hawk, we have better control over this nonaxisymmetry, which simplifies modeling.

The second goal of this paper is to investigate the role of stochastic ion acceleration in DPFs. Stochastic acceleration is a process where the charged particles scatter around plasma flow, and loose and gain energy mostly due to motional electric fields $\mathbf{E} = [\mathbf{v} \times \mathbf{B}]/c$. This results in the particle distribution developing a power-law tail with a small fraction of particles

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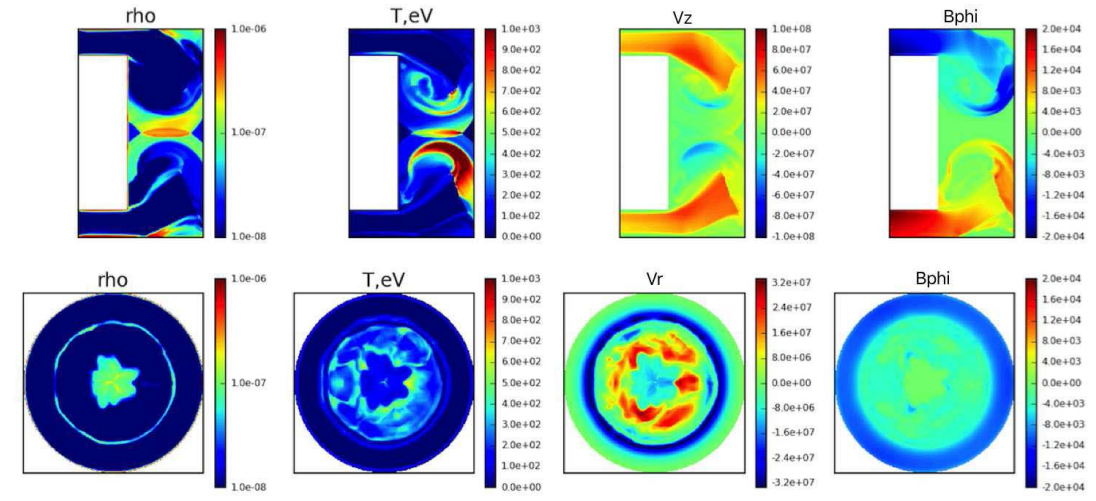


Fig. 1. Several dynamical quantities from a 3-D simulation with initial density having an $m = 3$ perturbation. We stopped the simulation during the first stagnation on the axis (pinch formation) at around $1 \mu\text{s}$. The top row depicts datacube sliced along the symmetry axis, while the bottom slices the datacube through the middle of the device, perpendicular to the axis. All units are CGS, except temperature, which is in electronvolt. Note that despite the initial perturbation and the instabilities which make the flow strongly nonaxisymmetric, the maximum density and temperature were still obtained close to the axis.

gaining significant energies [7]. This mechanism is especially efficient when shocks are present. The plasma dynamics in a DPF is dominated by shocks and shear flows, and particles can travel upstream and downstream of the shock gaining energy multiple times. We hypothesize that stochastic acceleration may play a role in providing initial acceleration of ions for injection into the current disruption region, where they undergo further acceleration by the nonideal electric field.

In the following, we describe our effort to carry out the 3-D MHD modeling of the pinch formation in the Hawk DPF geometry, as well as test particle simulations where particles were evolved in the ideal E and B fields from the simulation. We did not yet investigate the final stage of acceleration due to nonideal fields, which will be the subject of future work.

II. SIMULATION SETUP

We used Athena, a code developed in Princeton [8]. It is a finite volume MHD code based on approximate Riemann solvers. It uses an Harten–Lax–van Leer–Discontinuities or Harten, Lax, van Leer and Einfeldt scheme with up to third-order reconstruction in space and second order in time. Nonideal terms include viscosity, resistivity, Hall, and ambipolar diffusion terms. It is publicly available on GitHub [9]. Some of our ideal MHD simulations have been performed with a newer, faster version of Athena called Athena++, also available on GitHub [10].

An equivalent circuit was used to model the Hawk pulsed-power generator. The simulated DPF plasma device was connected in series with a 720-nH inductance, 0.15- Ω resistance, and 1.07- μF capacitance. The initial state of this circuit was zero current and 640-kV voltage on the capacitor.

The MHD simulation was coupled to the circuit in the following way: the input to the simulation from the circuit was the current I and current derivative in time dI/dt , which determined two boundary conditions on the open end of the simulation domain: 1) the current determined the inflow azimuthal magnetic field B_ϕ by Biot–Savart’s law and

2) dI/dt was used to determine the inflow velocity gradient in the z -direction by Faraday’s law. All other velocities and magnetic field components were set to zero. It is worth noting that the velocity gradient in the z -direction played a negligible role in the dynamics except for at the beginning of the acceleration phase. Setting this gradient to zero affected the results only negligibly. The output of the simulation and input to the circuit were the voltage on the DPF device, which was directly calculated by integrating the electric field across the terminals on the inflow entrance. The Athena MHD code required a nonzero inflow density of the plasma, which we typically set to between 1/10th and 1/100th of the plasma density inside the device.

The ordinary differential equations describing the circuit evolution was evolved using the first-order Euler method, as a subroutine of the boundary condition routine in Athena, with a time step equal to the time step chosen by Athena. Given that the characteristic timescale of the circuit was a microsecond, while the Athena time step was typically below 0.1 ns, such a circuit solver was sufficiently precise for our purposes.

We used the geometry approximating the actual DPF installed on Hawk, with anode radius of 6.33 cm and anode length of 4 cm, and cathode radius of 8.57 cm. We tried to model the plasma injection by plasma guns with fairly complex initial density distribution: 1) one component was placed between the anode and the cathode and modulated by a sine wave in the azimuthal angle ϕ , with the ratio of the high-to-low density of a factor of 2, the average density of this component we refer to as fill density ρ_0 and 2) the rest of the device was filled uniformly with density of 1/4 of ρ_0 . The azimuthal modulation had cases with uniform distribution, $m = 0$ and perturbations with mode numbers $m = 3$ and 6. This distribution was approximated from the available measurements done in the device after the firing of plasma guns. We assumed plasma in fully ionized deuterium, so that a characteristic density of 10^{-7} g/cc would correspond to the number density of $3 \times 10^{16} \text{ cm}^{-3}$. The number density was

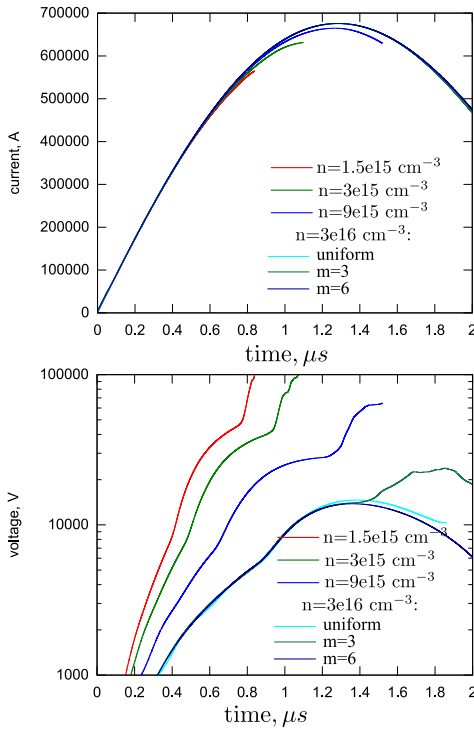


Fig. 2. Current and voltage versus time in simulations with a different fill density. The voltage is measured at the entrance of the DPF device, i.e., simulation domain. The DPF device has zero voltage at the beginning of the simulation, because at $t = 0$ we inject well-conductive plasma near the terminals of the device. At $t = 0$ plasma is not moving, so the inductive voltage is zero. Resistive voltage is close to zero as well due to the low resistivity of the plasma.

optimized to have the maximum current during the pinch. We used Lee's model [11] to estimate the pinch time and chose density around this value. These density values, however, are smaller than those used in DPFs filled with neutral gas [12], [13]. The deviation between the fill density and the plasma density is included in Lee's model as a parameter, presumably because only a fraction of neutral gas is being ionized and swept toward the pinch. We assumed this parameter around 0.2, i.e., our plasma density will be equivalent to fill pressure that is five times higher. Our initial guess for the plasma density was mostly correct as evidenced in Fig. 2, where the pinch time for the $3 \times 10^{16} \text{ cm}^{-3}$ density is $0.95 \mu\text{s}$ that is close to the peak of the current.

Another important parameter affecting neutron production in DPFs is the so-called drive parameter, related with the plasma velocity, $I_{\text{max}}/ap^{1/2}$, where I_{max} is the current during pinch, a is the anode radius, and p is the equivalent fill pressure in mbar [13], [14]. For the plasma density of $3 \times 10^{16} \text{ cm}^{-3}$, this parameter is around $63 \text{ kA/cm/mbar}^{1/2}$, which is close to the optimal value of $70 \text{ kA/cm/mbar}^{1/2}$ in [13] and [14].

Most of the 3-D simulations were performed with grid resolution of $n_x \times n_y \times n_z = 480^2 \times 288$.

III. MHD RESULTS

Fig. 1 presents a simulation at the moment of pinch formation for illustrative purposes. We plot 2-D slices of the 3-D data as two cuts one along the axis of the device (top row)

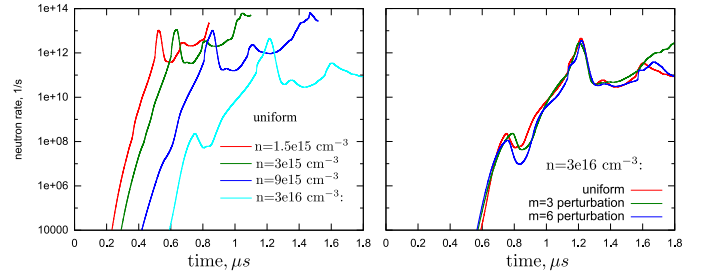


Fig. 3. Thermal neutron fluence from 3-D ideal MHD simulations. Neutron fluence for different initial fill densities (Left). Neutron fluence for several different axial mode perturbations (Right).

and another perpendicular to the axis in the middle of the device (bottom row). The density, temperature, axial plasma velocity, and azimuthal magnetic field are shown in color.

We observe that the evolution of the collapse after the shock turns around the corner of the anode becomes fairly unstable and develops perturbations not only at $m = 3$ but at higher modes as well. At the same time, the instabilities in the axial plane do not emerge until after the collapse.

Two of the quantities measured in the DPF experiment are the current through the device and the voltage output by the generator. The evolution of current and voltage is shown in Fig. 2, using a series of simulations with different fill densities and different initial azimuthal perturbations. Our target experimental density is $3 \times 10^{16} \text{ cm}^{-3}$. At this density, the voltage on the device is rather low, typically below 10 kV, so that the current does not vary significantly from the value expected from a short circuit, which is a sine wave with a period of $5.2 \mu\text{s}$. The voltage behavior has two components: voltage grows quickly until the pinch occurs. Afterward, the growth slows down and the second spike occurs when the plasma mostly leaves the device.

To study the nonaxisymmetric effects on the pinch formation, we have to introduce a metric of the pinch collapse by combining the density and the temperature. A useful combination of this sort is the thermal neutron yield, this also gives us an estimate of the total thermal fusion yield. Note that the thermal fusion is thought to be subdominant in DPFs, so these numbers should not be taken as actually projected yields for the Hawk DPF.

Fig. 3 shows the fluence of thermal fusion neutrons versus time for different initial plasma densities as well as for different types of azimuthal perturbations. The figure shows that the amplitude of the initial neutron pulse increases with decreasing density, but its duration decreases. While the neutron rate only increased with decreased density, the total yield, as a function of initial density, was found to have a maximum at $n = 9 \times 10^{15} \text{ cm}^{-3}$. Introducing initial density perturbations did not prevent the collapse on axis in these simulations. These simulations also suggest that the collapse quality is about the same for the $m = 3$ and $m = 6$ perturbations that are two viable choices for the Hawk DPF design realized with three and six plasma guns correspondingly.

We studied the effects of the nonideal terms in the induction equation by introducing the Hall term as well as collisional resistivity, both of which are calculated using the actual plasma

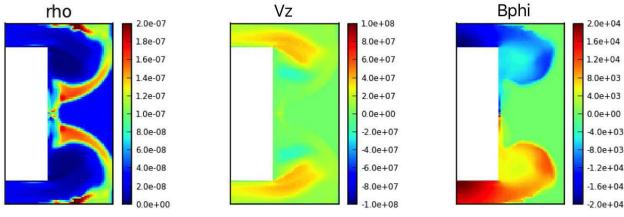


Fig. 4. Result of a nonideal Hall-MHD simulation with the positive polarity. The column is pinched close to the anode. The nonideal terms included in this simulation were the Hall term and Spitzer collisional resistivity.

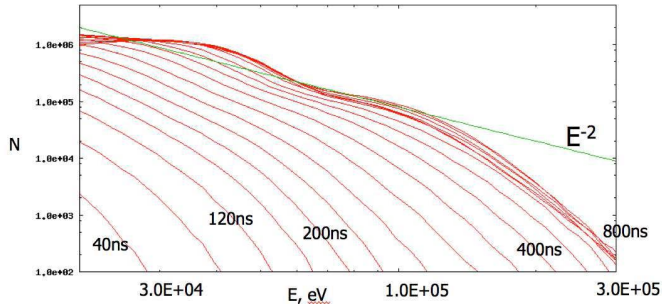


Fig. 5. Spectrum of particles resulting from the stochastic acceleration of ions by the ideal electric field.

density and temperature. The resistivity we used was from the standard collisional formula, i.e., “Spitzer resistivity” [15]. Note that the Hall term introduces 3-D effects by breaking the mirror symmetry of MHD equations. Thus, the effect of adding this term should be different for “positive polarity” where the center conductor is the anode, and “negative polarity” where the center conductor is the cathode. The dynamical quantities ρ , v_z , and B_ϕ are shown in Fig. 4 at the time the pinch first reaches the axis. After the shock turned around the anode corner and started pinching axially, the effect of the Hall term was visible in that the flow of electrons toward the anode pushed the magnetic field closer to the anode, and resulted in a faster and tighter pinch close to the anode end. This is qualitatively in agreement with the experimental data and particle-in-cell simulations where the pinch disruption occurs close to the anode [4], [16]. The introduction of resistivity did not seem to affect the dynamics qualitatively.

IV. PARTICLE SIMULATIONS

Hephaestus is parallelized, multipurpose, particle in cell-MHD code with stochastic terms and particle splitting and merging [17]. We used Hephaestus to simulate test particles propagating in the ideal MHD fields in the DPF device as the fields were evolving. The particles were continuously injected into the simulation domain with an isotropic Maxwellian distribution at 1-keV temperature. Note that the maximum plasma temperature in our simulations was reached during the pinch formation and was around 3 keV, although in most of the volume the temperature was much lower. We also assumed Bohm scattering, which corresponds to the efficient self-scattering of ions due to streaming instabilities. The result is presented in Fig. 5. A smaller portion were accelerated by a stochastic mechanism to a power-law tail extending to

around 200 keV. This energy, 200 keV, is consistent with the cutoff for the stochastic mechanism that would be expected in this system. This cutoff energy is estimated by assuming that particles at this energy have Larmor radii of the order of the characteristic size. Note that such estimate is independent on the distance from the axis. If the characteristic distance is the distance to the axis R then, since the magnetic field also scales as $1/R$, the Larmor radius will scale as R and the ratio of the Larmor radius to R does not depend on R .

V. CONCLUSION

We performed 3-D MHD simulations with the finite volume code Athena. We implemented a simplified circuit model of the generator, and coupled it to Athena to drive the current in the simulation. We also modeled the nonaxisymmetric plasma injection that might be expected with the plasma guns used in the experiment. We noticed only a small difference in the collapse quality between a uniform initial plasma case, and simulations with perturbations setup to model three and six guns. The plasma collapse in three dimensions was never perfectly axisymmetric due to instabilities, perturbations in the initial density, and the grid perturbations. The thermal neutron count was also lower than predicted by simplistic models with perfect collapse.

Since conventional wisdom in DPFs is that the accelerated particles likely dominate the neutron yield through the beam-target fusion reactions. We thus investigated a stochastic acceleration mechanism, where ions receive energy by crossing shocks multiple times and accelerate in ideal (motional) electric fields [7]. The maximum energy of such acceleration was around 200 keV, which is about a factor of 3 lower than the generator voltage, and thus less than the expected current disruption voltages. Perhaps more importantly, the stochastic acceleration will result in the largely isotropic distribution of ions, while in DPFs the high-energy tail of ions is usually a beam parallel to the axis. The stochastic acceleration of ions could, however, work as a mechanism to inject high-energy particles into the final stage of acceleration, where nonideal fields accelerate particles during the current disruption. Our MHD simulations at this point cannot model the current disruption because it neglects the nonideal physics that becomes important during the disruption.

We also started to investigate the effect of nonideal physics. In the case of the Hall term, it was found to affect the dynamics close to the anode, resulting in a pinch close to the anode. This effect is only observed in the positive polarity case, and was not seen in negative polarity. This qualitatively agrees with experimental observations that the current disruption occurs near the anode [4], [16].

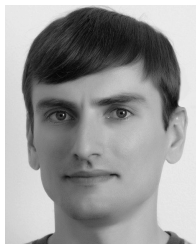
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Dr. Jackson is the Treasurer for the 2019 IEEE Pulsed Power and Plasma Science Conference, a combination of the 22nd IEEE Pulsed Power Conference, and the 46th IEEE International Conference on Plasma Science.

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